Hands-on Activity No. 6.1

Hands-on Activity 6.1: Function Basics

Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming logic and Design	**Date Performed:September 3, 2026**
Section: CPE11S2	**Date Submitted: September 3, 2026**
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1. Objective(s)

This activity aims to develop programming skills using functions.

2. Intended Learning Outcomes

2.1 Create a program in C++ that will add, subtract, divide, multiply for a 2 user-input integer values. Use functions in the program.

2.2 Create a program in C++ that will convert a user-input Farenheight to Celcius and vice versa. Use functions in the program.

2.3 Create a program in C++ that will convert dollars into pesos and vice versa. Use functions in the program.

3. Materials and Equipment

Internet connection
Personal Computer

4. Output

2.1 Create a program in C++ that will add, subtract, divide, multiply for a 2 user-input integer values. Use functions in the program.

Answer:

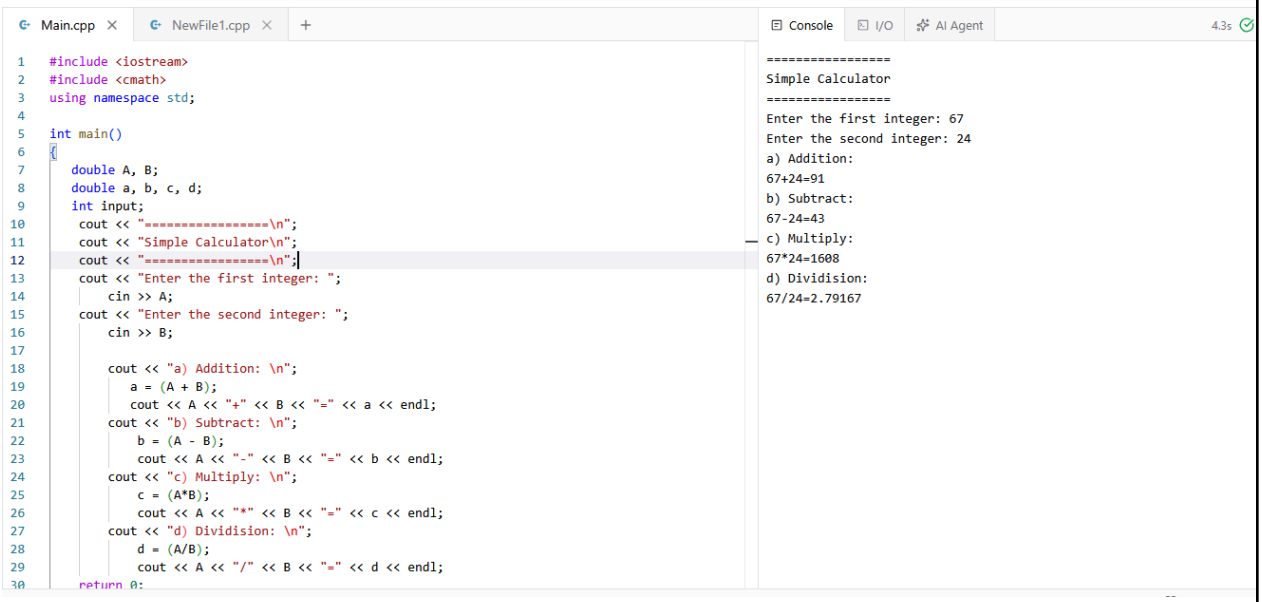
```
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
   double A, B;
   double a, b, c, d;
   int input;
    cout << "=================\n";
    cout << "Simple Calculator\n";
    cout << "=================\n";
    cout << "Enter the first integer: ";
        cin >> A;
    cout << "Enter the second integer: ";
        cin >> B;

        cout << "a) Addition: \n";
           a = (A + B);
           cout << A << "+" << B << "=" << a << endl;
```

```
        cout << "b) Subtract: \n";
            b = (A - B);
            cout << A << "-" << B << "=" << b << endl;
        cout << "c) Multiply: \n";
            c = (A*B);
            cout << A << "*" << B << "=" << c << endl;
        cout << "d) Dividision: \n";
            d = (A/B);
            cout << A << "/" << B << "=" << d << endl;
    return 0;
}
```

ScreenShot:



2.2 Create a program in C++ that will convert a user-input Fahrenheit to Celsius and vice versa. Use functions in the program.

Answer:

```
#include <iostream>
using namespace std;

int main() {
    double Ftemp, Ctemp;
    double F, C;
    int value;
    int ans;
        do { //Used a Do-while loop so that the code continues until the
user chooses to end the program
```

```
        cout << "----------------------" << endl;
        cout << "Temperature Conversion" << endl;
        cout << "----------------------" << endl;

        cout << "Celsius to Fahrenheit(1)" << endl;
        cout << "Fahrenheit to Celsius(2)" << endl;
        cout << "Exit Program(3)" << endl;

        cout << "----------------------" << endl;
        cout << "Select Option (Type Answer):" << endl; // The selection
menu of the conversion
        cin >> ans;
    if (ans == 1) { //An if for a condition to be able to choose between
the option
        cout << "Enter Temperature in Celsius: " << endl;
        cin >> Ctemp;
        F = (Ctemp * 9/5) + 32;
        cout << "Celsius in Fahrenheit is: " << F << endl;
    }
    if (ans == 2) {
        cout << "Enter Temperature in Fahrenheit: " << endl;
        cin >> Ftemp;
        C = (Ftemp - 32) * 5/9;
        cout << "Celsius in Fahrenheit is: " << C << endl;
    }
    }
    while (ans!=3); //as long as it is not equal to 3 the program will not
end
        cout << "Thank you for using the conversion!" << endl;
        cout << "Ending Program" << endl;
    return 0;
    }
```

Screenshot:

```cpp
#include <iostream>
using namespace std;

int main() {
    double Ftemp, Ctemp;
    double F, C;
    int value;
    int ans;
        do { //Used a Do-while loop so that the code continues until the user chooses to end the
        cout << "----------------------" << endl;
        cout << "Temperature Conversion" << endl;
        cout << "----------------------" << endl;

        cout << "Celsius to Fahrenheit(1)" << endl;
        cout << "Fahrenheit to Celsius(2)" << endl;
        cout << "Exit Program(3)" << endl;

        cout << "----------------------" << endl;
        cout << "Select Option (Type Answer):" << endl; // The selection menu of the conversion
        cin >> ans;
    if (ans == 1) { //An if for a condition to be able to choose between the option
        cout << "Enter Temperature in Celsius: " << endl;
        cin >> Ctemp;
        F = (Ctemp * 9/5) + 32;
        cout << "Celsius in Fahrenheit is: " << F << endl;
    }
    if (ans == 2) {
        cout << "Enter Temperature in Fahrenheit: " << endl;
        cin >> Ftemp;
        C = (Ftemp - 32) * 5/9;
```

```
----------------------
Temperature Conversion
----------------------
Celsius to Fahrenheit(1)
Fahrenheit to Celsius(2)
Exit Program(3)
----------------------
Select Option (Type Answer):
2
Enter Temperature in Fahrenheit:
43
Celsius in Fahrenheit is: 6.11111
----------------------
Temperature Conversion
----------------------
Celsius to Fahrenheit(1)
Fahrenheit to Celsius(2)
Exit Program(3)
----------------------
Select Option (Type Answer):
1
Enter Temperature in Celsius:
43
Celsius in Fahrenheit is: 109.4
----------------------
Temperature Conversion
----------------------
```

2.3 Create a program in C++ that will convert dollars into pesos and vice versa. Use functions in the program.

Answer:

```cpp
#include <iostream>
using namespace std;

int main() {
    const double RATE = 63.06;  // PHP per 1 USD; update as needed
    int choice;
    double amount, result;

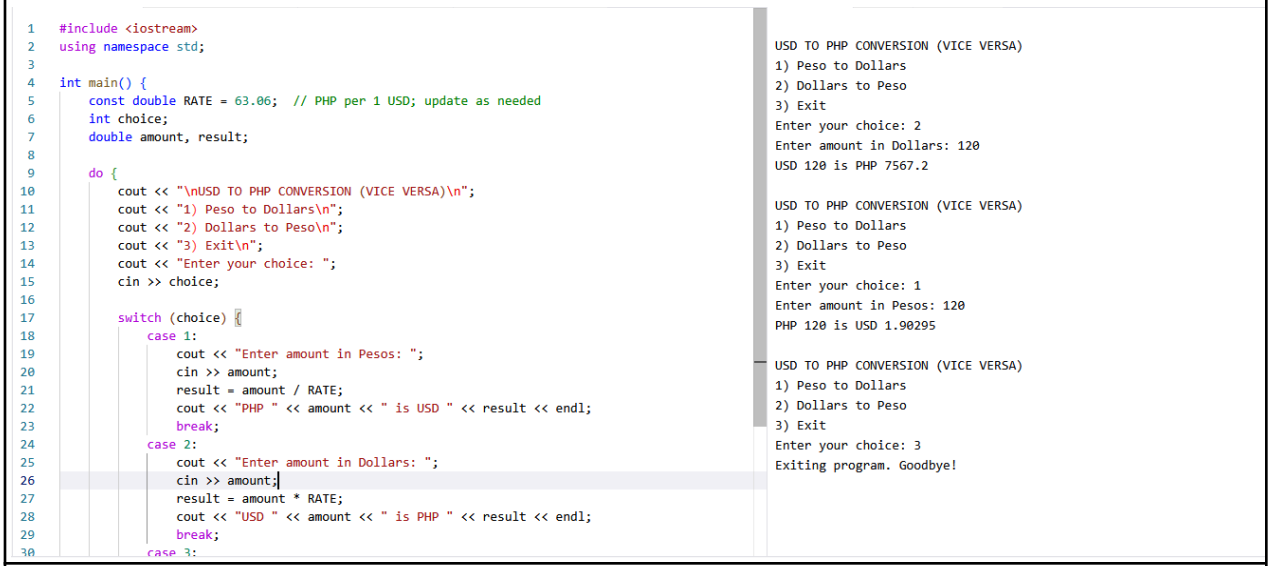
    do {
        cout << "\nUSD TO PHP CONVERSION (VICE VERSA)\n";
        cout << "1) Peso to Dollars\n";
        cout << "2) Dollars to Peso\n";
        cout << "3) Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                cout << "Enter amount in Pesos: ";
                cin >> amount;
                result = amount / RATE;
                cout << "PHP " << amount << " is USD " << result << endl;
                break;
            case 2:
                cout << "Enter amount in Dollars: ";
```

```
                cin >> amount;
                result = amount * RATE;
                cout << "USD " << amount << " is PHP " << result << endl;
                break;
            case 3:
                cout << "Exiting program. Goodbye!" << endl;
                break;
            default:
                cout << "Invalid choice. Please enter 1, 2, or 3." <<
endl;
        }
    } while (choice != 3);

    return 0;
}
```

Screenshot:



```
#include <iostream>
using namespace std;

int main() {
    const double RATE = 63.06;  // PHP per 1 USD; update as needed
    int choice;
    double amount, result;

    do {
        cout << "\nUSD TO PHP CONVERSION (VICE VERSA)\n";
        cout << "1) Peso to Dollars\n";
        cout << "2) Dollars to Peso\n";
        cout << "3) Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                cout << "Enter amount in Pesos: ";
                cin >> amount;
                result = amount / RATE;
                cout << "PHP " << amount << " is USD " << result << endl;
                break;
            case 2:
                cout << "Enter amount in Dollars: ";
                cin >> amount;
                result = amount * RATE;
                cout << "USD " << amount << " is PHP " << result << endl;
                break;
            case 3:
```

```
USD TO PHP CONVERSION (VICE VERSA)
1) Peso to Dollars
2) Dollars to Peso
3) Exit
Enter your choice: 2
Enter amount in Dollars: 120
USD 120 is PHP 7567.2

USD TO PHP CONVERSION (VICE VERSA)
1) Peso to Dollars
2) Dollars to Peso
3) Exit
Enter your choice: 1
Enter amount in Pesos: 120
PHP 120 is USD 1.90295

USD TO PHP CONVERSION (VICE VERSA)
1) Peso to Dollars
2) Dollars to Peso
3) Exit
Enter your choice: 3
Exiting program. Goodbye!
```

5. Conclusion

Deeper Understanding of Functions

6. Rubric

SO7(14)

Rubric for SO 7 (14)

Criteria	Ratings					
SO 7 PI 1 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or i acquiring new knowledge
SO 7 PI 2 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to co independently
SO 7 PI 3 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather Informat variety of sources
SO 7 PI 4 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied from the sources consulted